Hedonic Model, Random Forest and Artificial Neural Network: Comparison for Real Estate Price Prediction in Lisbon

Abstract

The real estate market is a complex industry with many variables affecting the prices of transactions. Formerly, understanding this market could be restricted by the lack of data or tools. However, with the current developments in the Artificial Intelligence field, three machine learning models (Hedonic Pricing Regression Model, Random Forest and Artificial Neural Network) are deployed and set side by side on the topic of predicting the price per square meter of housing properties, using real-estate data from Lisbon, Portugal. Overall, model evaluation reveals that the Random Forest was the best performing model, followed by the Artificial Neural Network and the Hedonic Model. These can predict the selling price per square meter accurately and learn the importance of certain features. Furthermore, the models implemented allow for users in the real estate paradigm to accurately and objectively assess the selling price or asking price of a dwelling, as well as estimate the valuation of certain features of a property.

Keywords

Real Estate, House Price Prediction, Machine Learning (ML), Hedonic Pricing Model (HPM), Random Forest (RF), Artificial Neural Network (ANN)

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1. Introduction

The real estate financial market is a complex supply chain, and it is prone to a lot of changes, whether due to endogenous factors that have to do with the market itself - such as pricing of the sales/rent and the number of houses available for sale/rent - but also due to exogeneous factors, that are external to the real estate market, such as political, economic and demographic (Giannotti, 2008). In the past, understanding the behavior of this market could be limited by the lack of data, since it couldn't capture the highly rich networks that exist nowadays between the financial institutions and the financial products. However, with the continuous growth of data available in the real estate market, it is now possible to use different disciplines, such as machine learning and statistics, to turn raw information into value (Van der Aalst, 2016). Many changes have come to the real estate market, allowing that further market research can be performed in subjects, such as predictive modelling, natural language processing or geospatial analytics (Bekkerman et al., 2020). The subject matter of this study, the capital of Portugal, Lisbon, is showing tremendous evidence that point out a significant growth on tourism and real estate investment, since it is viewed as a great place to live for tourists or international students, that are captivated by the great weather, inexpensive costs of living, landscape sites, safety and cosmopolitan leisure (Santos, 2019). Following the 2008 financial crisis, the Portuguese government has issued state measures, in order to captivate foreign investment, such as low personal taxation to EU citizens in 2009, or the Golden Visa program in 2012, which allowed for non-EU citizens who make investments in the real estate to have a Portuguese citizenship and have access to the Schengen area (Cocola-Gant & Gago, 2021). Due to these measures, the gentrification of the urban areas of Lisbon is a recurring topic that has affected the real estate pricing, since several structural modifications are being accommodated such as the rehabilitation of the properties and has ultimately transformed Lisbon inhabitant's paradigm (Krähmer & Santangelo, 2018). Over and above that, in 2022, there are many variables that are contributing for the rise in prices in Portugal and Europe, such as the climb of inflation rates and interest rates, as well as the disruption of supply chains caused by the armed conflict in Europe or even by the reminiscences of COVID-19 (Linhart et al., 2022). According to Instituto Nacional de Estatística, in the last trimester of 2021 the median price of family housing facilities in Portugal was equal to 1355€/m^2 per square meter, registering a staggering ascent of 14,1% when compared to the homologous period (Instituto Nacional de Estatística, 2022). However, in the last trimester of 2021 all the parishes in the Metropolitan Area of Lisbon recorded higher prices than the national median price, reaching up to 3723€/m^2 and making it one of the most expensive regions in the country (Instituto Nacional de Estatística, 2022). Therefore, it is compelling to analyze this market using state of the art technologies, not only because housing accommodations represent a tremendous role in global economy and are an important part of everyone's life, but also because the real estate data keeps on getting more and more available and it is fundamental to adapt such vast amounts of data to the machine learning paradigm (Renigier-Biłozor et al., 2022). In short, the real estate market needs analytic and predictive technologies that could potentiate the good usage of the existing data and the ever-growing amount of artificial intelligence technologies, in order to extract the maximum of the previous mentioned tools. On that account, with the data gathered from an independent databank that produces information systems about real estate in Portugal, the performances of three different regression models commonly used in the related literature - Hedonic Model, Random Forest Model and Artificial Neural Network - are analyzed and evaluated, to investigate what is the optimal approach to predict the transaction price of a housing facility. This paper is organized as follows: Section 2 describes the related works that have been developed in this thematic, Section 3 illustrates the methodology that was followed to produce the results obtained, Section 4 describes the results obtained and Section 5 provides the conclusions.

2. Literature Review

2.1. Hedonic Pricing Models

Hedonic Pricing Models (HPM) were introduced by Lancaster in 1966 and were later applied to the real estate market by Rosen in 1974, in order to assess the valuation of real estate, since housing properties are not homogeneous goods and are influenced by a variety of intrinsic characteristics (Lancaster, 1966; Rosen, 1974). In their work, Sirmans et al. studied the interaction between structural, geographical and environmental

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housing variables (Sirmans et al., 2005). The top characteristics used to specify hedonic pricing equations and compare coefficient estimates by geographical area were also analyzed, showing that some variables have a negative effect on the selling price (Sirmans et al., 2005). Similarly, the performance of the hedonic model in a set of different scenarios is studied, with different housing facilities in different cities and concluding that these models are useful in order to predict future transaction prices, as well as to determine the valuation of different house characteristics, independently of the location (Monson, 2009). For example, in (Bottero et al., 2018) works, the interaction of spatial effects in estimating hedonic model for buildings energy efficiency in the urban area of Turin, Italy was analyzed. The main purpose of the usage of HPM in this research is to estimate implicit marginal prices as a measure of willingness to pay for buildings energy performance, as there is a need to develop incentive policies for improving buildings energy efficiency.

2.2. Artificial Intelligence Models

Due to the increase in computational processing in the past few years, several Machine Learning techniques are being applied to the real estate price prediction theme. Nowadays, tree-based approaches (Fan et al., 2006, Ho et al., 2021), Artificial Neural Networks (Rampini & Cecconi, 2021; Peter et al., 2020) and other machine learning models (Park & Bae, 2015; Madhuri et al. 2019) are being used in order to surpass the limitations imposed by the Hedonic Model. In this research, a Random Forest and Artificial Neural Networks (ANN) are compared, with the main advantages of using these algorithms being their ability to deal with complex and noisy datasets, and finding unknown relationships in the data (Georgiadis, 2018).

2.2.1. Random Forest

The Random Forest (RF) algorithm was proposed in 2001, and it is a supervised machine learning technique that makes use of ensemble learning (Breiman, 2001). Similarly to other machine learning models, it has been applied to the real estate market to estimate the housing valuation. There are several articles that study the behavior of the RF applied to the real estate market. Various classification models are compared in (Yu & Wu, 2016) work, such as Naïve Bayes, Support Vector Machine (SVM) Classification and RF, in the housing price prediction theme. The result was that the RF presented a Root Mean Squared Error (RMSE) lower than the baseline model and presented the second highest accuracy overall. Similarly, (Mohd et al., 2019) studied the utilization of machine learning algorithms for prediction house selling prices in Malaysia. Comparing the different algorithms, such as RF, Decision Tree, Ridge, Lasso and Linear Regression, the RF algorithm exhibits the best accuracy and the lowest RMSE. Furthermore, this machine learning algorithm was also studied, in order to build house price estimation models, obtaining similar results, and finding that it can capture nonlinear hidden relationships between house prices and house locations (Levantesi & Piscopo, 2020; Wang and Wu, 2018).

2.2.2. Artificial Neural Networks

Artificial Neural Networks (ANNs) were firstly introduced by the neuropsychologist Warren McCulloch and the mathematician Walter Pitts in 1943, which has been viewed as fundamental to the Artificial Intelligence paradigm (McCulloch & Pitts, 1943). Furthermore, (Noriega, 2005) explains that ANNs attempt to model the functioning of the human brain, that is constituted by billions of individual cells called neurons. More recently, ANNs have been applied to the real estate price prediction, as it can overcome some difficulties imposed by other models (Georgiadis, 2018). There are several works that study the performances of different algorithms, namely between performances of Hedonic Price Models and Artificial Neural Networks. For instance, Rampini and Cecconi, provide a comparison between the most popular evaluation method of properties, Hedonic Price Model, and some popular ML techniques, namely Artificial Neural Networks (Rampini & Cecconi, 2021). In their work, the real estate data is gathered from two cities in Italy, and the ANN was the ML technique which revealed the lowest Mean Average Error (MAE), proving that not only Machine Learning techniques work better with rich datasets, in order to train the models, but also presented the best results overall (Rampini & Cecconi, 2021). A Geographical Information System (GIS) based representation of the ANNs error helps prove that the network was able to learn the importance of the location in the training phase. Likewise, several models were studied and the

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advantage of the Artificial Neural Network (ANN) over the classical Hedonic Model was clear, showing that not only it presented more precise estimates of the transaction prices of the houses, but also presented better results for the marginal prices associated to each one of the characteristics of the corresponding property (Abidoye & Chan, 2018; Tabales et al., 2013; Peterson & Flanagan, 2009). However, the main downside about the ANN models is the interpretability of the associations made by the machine learning algorithm. In other words, the model may use features that are considered irrelevant to task, in order improve results, and the user won't understand the usage of those variables, since ANN's are a black-box system (Spiegelhalter, 2019).

3. Methodology

For the first part of the research, recurring to analytical, statistical and graphical techniques, an initial analysis of the data used for this research is made, in order to explore it, as well as to check the data quality (Schröer et al., 2021). Next, Data Preparation procedures are applied to the project. At this phase, the data should be submitted to data preprocessing techniques, such as data cleaning, data transformation, data selection and data imputation (this phase is further explained in Section 3.1). Having completed this phase, the data modelling stage begins, where the pipeline of model selection, test design, model building, and assessment is developed. As mentioned before, the models selected for this problem are: Hedonic Pricing Model Regression, Random Forest and Artificial Neural Network, and for each one a different pipeline is created, such that the maximum performance of each model can be obtained. For the test design part of this project, the dataset is split into two parts, the training set and the test set, following a distribution of 80% of the original data for the training data and 20% for the test data. The implementation of the models is made recurring to Python environments, using the Sklearn[1] library for the Hedonic Regression and Random Forest. For the Artificial Neural Network, the Keras[2] and TensorFlow[3] libraries are used. Applying Hyperparameter Tuning and other techniques, the corresponding hyperparameters are set for each one of the models. Lastly, the final results are evaluated and interpreted against the objectives. Moreover, all the previous steps are revised in order to check if there is margin for improvement (this phase is further explained in Section 4).

3.1. Dataset Description and Modifications

The real estate data comes in three broad types: (1) financial data, which includes information about Real Estate Investment Trusts (REITs) shares and related stocks; (2) transactional data, which refers to the to the information on the real estate purchases, expenses or taxes; (3) physical data, which includes information about the real estate property itself, such as structural characteristics or locational data (Winson-Geideman & Krause, 2016). The data used for this research focus mainly on the transactional and physical data of the properties, having information about the properties that are listed for sale and the sales already executed, and makes use of the physical data in order to assess the price. Initially, temporal and locational filters are applied in order to choose the data points that are ranging from 2017 to 2022 and are located for the Metropolitan Area of Lisbon in Portugal. Afterwards, data engineering techniques had to be exerted in order to proceed to the modelling phase, namely checking for duplicated observations, converting data types, checking and removing outliers. In addition to this iterative data engineering process, and since this dataset had an extensive number of missing values in some variables, data imputation procedures were applied, such as KNN Imputation, Logistic and Linear Regression Imputation and Decision Tree Imputation. For the Regression imputations, and as the variables Pool, Terrace, Patio and Garage are binary variables, Logistic Regression Imputation was used to complete these variables. For the Energy Efficiency Certificate (EEC), a polynomial variable, Linear Regression Imputation was applied. In the end, the imputation method chosen was the KNN Imputer, with nearest neighbors=3, since the variables created by this imputer produced the best results (further explained in the section 4) when compared to the other imputation algorithms. Lastly, other important variables are created from the existing ones, extracting years from dates,

[1] https://scikit-learn.org/stable/
[2] https://keras.io/
[3] https://www.tensorflow.org/

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transforming observations into polynomial variables, and creating the dependent variable, that, for this research, is the price in euros in function of the size of the property (Price €/m^2). Moreover, when dealing with regression algorithms it is important to perform scaling on the numerical data, since normalizing the data can improve the predictability of the machine learning algorithms (Nayak et al. 2014). In the end, a clean, simple and effective dataset, with 15 variables and around 37.000 observations is generated and ready to be applied for the modelling phase. In this dataset, the properties that have already been sold have the final listing price, which is used to output the dependent variable. The final dataset, after all the operations described above is described in Table 1.

	Description	Variable Type
Property Type	Indicator if an apartment or a house (0=flat, 1=house)	Binary
Number of bedrooms	Number of bedrooms in a property	Numerical
Total Gross Area	Total gross area of the property in square meters	Numerical
Latitude	Latitude coordinate	Numerical
Longitude	Longitude coordinate	Numerical
Num. Conservation	Conservation state of the property	Numerical
Bool. Conservation	Conservation state information is obtained from customers (0) or if obtained through images (1)	Binary
Year Initial Offer	Year of the Initial Offer for the property	Numerical
Pool	Boolean for existence of a Pool	Binary
Patio	Boolean for existence of a Patio	Binary
Garage	Boolean for existence of a Garage	Binary
Terrace	Boolean for existence of a Terrace	Binary
Energy Efficiency Certificate (EEC)	Energy efficiency of a property on a scale from A to G	Polynomial
Construction Year	Year in which the property was constructed	Numerical
Price/m2	Final Observed Price divided by the Total Gross Area	Numerical

Table 1 – Dataset variables.

3.2. Model Specification

This research compares the performance of three machine learning models (Hedonic Regression, Random Forest and Artificial Neural Networks), with different theoretical backgrounds, that output different evaluation scores, as well as model interpretations. This section attempts to explain these algorithms:

Hedonic Regression: In general, a hedonic equation is a regression equation that has its base on the housing characteristics, where the independent variables represent the individual characteristics of each one of the properties and the regression coefficients may be transferred into estimates of the implicit prices of these characteristics (Malpezzi et al., 2002). These attributes determine the value of the price index, and it may depend on characteristics such as structural characteristics, neighborhood characteristics, location within the market, contract conditions and even the time the property is listed in the market. The method of hedonic equations can be decomposed into measurable prices and quantities, so that either different or identical dwellings can be compared. Besides that, as different houses contain different characteristics, these may also be valued differently depending on the buyer of the property, suggesting that housing is a heterogeneous good (Malpezzi et al., 2002). However, the hedonic model generally takes this form:

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Market Price = f (Physical characteristics, Other Factors)

Random Forest (RF): The Random Forest algorithm is a supervised machine learning technique, that makes use of ensemble learning. In 2001, the Random Forest scheme was proposed, defined as the combination of tree predictors such that each tree depends on the values of a random vector sampled independently and with the same distribution for all trees in the forest (Breiman, 2001). The algorithm was influenced by some ideas that were described in Breiman's article, such as the random split selection proposed by Dietterich, the geometric feature selection of Amit and Geman and the "random subspace" method of Ho, which does a random selection of a subset of features to use to grow each tree (Dietterich, 1998; Amit & Geman, 1997; Ho, 1998). Later in 2004, the substantial gains in classification and regression accuracy were also studied, which can be achieved by using ensembles of trees, where each tree of the ensemble is grown in accordance with a random parameter, and the final predictions are obtained by aggregating over the ensemble (Breiman, 2004).

Artificial Neural Network (ANN): The Multilayer Perceptron (MLP) is described as a unique model of neural networks, that consists of a system of simple interconnected neurons or nodes, which are arranged in multiple computational layers (Gardner & Dorling, 1998). Generally, this model represents a nonlinear mapping between three primary components, an input data layer, the hidden layers and the output measure layers, where the nodes of the layers are connected by weights and outputs signals. The hidden layers have two essential processes, the weighted summation functions and the transformation functions, where both of these functions relate the value from the input data to the output value, which will give an estimation of the property value. The weighted summation function usually replicates the product between the input values and the weights associated to the input values, summed by all of the hidden nodes. On the other hand, transformation function relates the summation values of the hidden layers to the output variable. There is a wide variety of transformation functions, such as, linear functions, sigmoid functions or Gaussian functions, but the linear sigmoid function is usually preferred due to its non-linearity, continuity, monotonicity and continual differentiability properties (Pagourtzi et al., 2003).

3.3. Model Evaluation Metrics

There is a wide variety of available performance metrics used for research when it comes to compare performances of different machine learning regression models. Amongst others, the most used model evaluation metrics are RMSE, that represents the sample standard deviation of the differences between predicted and observed values, and MAE (Mean Absolute Error), that is a sum of the magnitudes of the errors divided by the total amount of observations. For this research, the preferred evaluation metric is the MAE, since it's a more natural measure of average error (Willmott & Matsuura, 2005). Consequently, a minor value of the metric MAE, means that the error is smaller, and the algorithm is precise. The formula for MAE is presented below:

$$\mathrm{MAE} = \frac{1}{n}\sum|y - \hat{y}|,$$

Moreover, MAPE (Mean Average Percentage Error), a measure based on percentage error, is also used to compare the performance of the models. A bigger value for MAPE means that the error is also bigger. The formula for MAPE is presented below:

$$\mathrm{MAPE} = \frac{100\,\%}{n}\sum\left|\frac{y - \hat{y}}{y}\right|,$$

Where:

$$y = actual\ output\ value,$$
$$\hat{y} = predicted\ output\ value,$$
$$n = number\ of\ data\ points$$

4. Results and Discussion

4.1. Exploratory Data Analysis

In Figure 1, the geographical distribution of the properties in function of the average price per square meter in Lisbon is represented by a hexagon, with an approximate length of 200 meters. The average final selling prices range from approximately 1014€/m^2 to 7934€/m^2, and it can be concluded that the most expensive properties are located in the center of the city or in the riverside areas. On the other hand, the least expansive areas are located in the least touristic areas of Lisbon, usually more distant from the center of city, where housing facilities have least conditions. The

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average price per square meter, throughout all the years being studied is approximately 3700€, and the standard deviation is approximately 1400€. Looking over the data, in the last two years, following the post-pandemic situation, the average price of the square meter in Lisbon rose to nearly 110%, since March 2020 until December 2021, and a staggering climb of 24% since the data for this research is available (since the beginning of the year 2017). Regarding the figure, it is important to point out that some regions don't have any information, because they are public domains where housing facilities are unavailable, such as public gardens, airports or monuments. Furthermore, there is also the possibility that the dataset doesn't have any information about that area in the time period that is being studied.

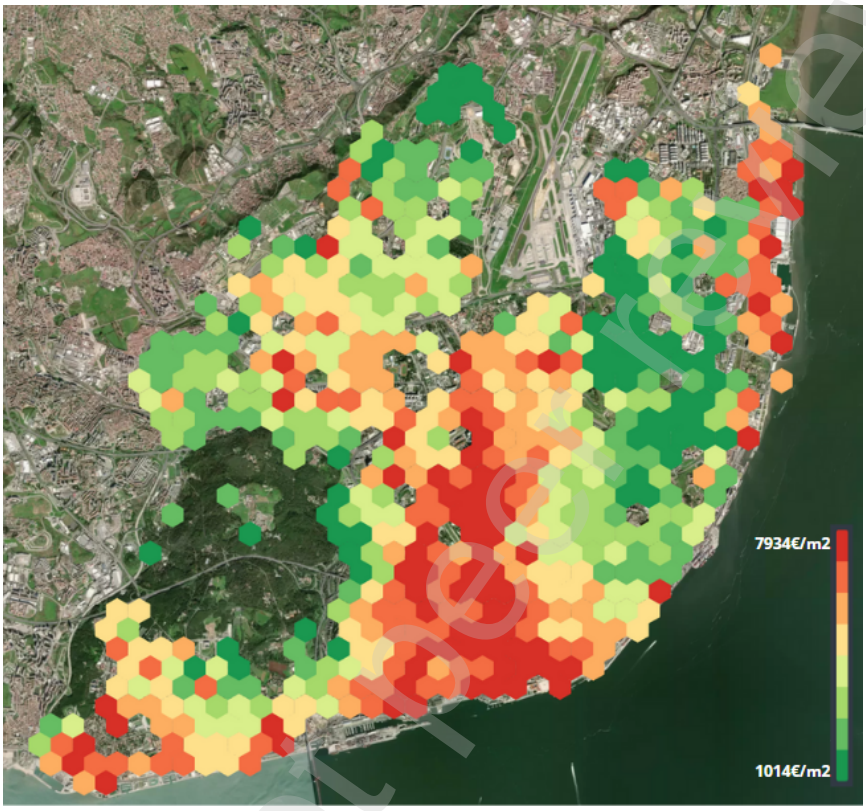

Figure 1 – Average of Observed Prices per Square Meter for the properties in Lisbon, Portugal.

4.2. Data Imputation Results

As mentioned in Chapter 3, Data Imputation techniques had to be applied in order to complete some variables. The imputation technique that registered the best overall results was the KNN Imputer, and therefore the variables imputed by this registered the best overall results was the KNN Imputer, and therefore the variables imputed by this method were the ones used in the final model. In Table 2, a comparison of the MAE Score for the different methods of imputation used is exhibited.

MAE Score	Linear Regression	Random Forest	Artificial Neural Network
KNN Imputer	985,267	623,645	789,674
Linear/Logistic Regression Imp. *	1000,053	627,075	814,050
Decision Tree Imp.	991,266	624,222	802,073

Table 2 – Comparison of the MAE for the different Data Imputation Techniques for all the models in this research. * Logistic Regression Imputation was used for the variables Pool, Terrace, Patio and Garage, and Linear Regression was used for the variable EEC.

4.3. Modelling Results

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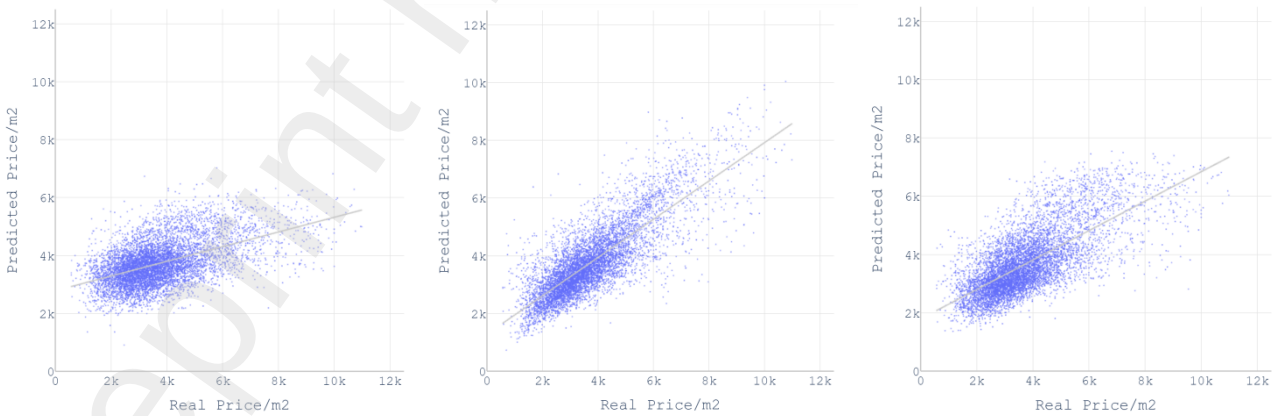
When comparing the performances of the models proposed for this research, both the MAE and MAPE scores are obtained from the test data which accounts for 20% of the dataset, after the original dataset modifications. Several combinations of the exogenous variables were experimented and checked against previous results, and the variables inputted that registered the best scoring metrics were the ones included in the final dataset. Table 3 shows the different performances of the algorithms, with the corresponding MAE and MAPE scores, after the data imputation variables were chosen. An examination of the previously mentioned table reveals that the Linear Regression model scored a MAE score equal to 985,297 €, demonstrating that the mean difference between the observed price and the predicted price is approximately 985 € per square meter. Similarly, the MAPE, that represents the average of the absolute percentage of the errors is equal to 30,628 %, values relatively high for the real estate price prediction paradigm. In figure 3, a scatterplot was plotted to confront the real price compared to the predicted price of each property, for all the models in this research. As it can be seen from figure 3a), the Linear Regression model predicts quite well the properties with a lower final price, while the predicted price for the properties with a higher final price were more far off. On the other hand, the Random Forest Regression performed quite well, registering better results than the Linear Regression, having a lower MAE = 623,645 € and MAPE = 19,386%, meaning that this algorithm is more accurate and is more likely to predict the price per square meter of the households with more precision. Figure 3b) also shows that the Random Forest not only was able to predict with more precision properties with a lower value, but also the ones with a higher value in Price per square meter. Likewise, for the Artificial Neural Network model, the scatterplot drawn in figure 3c), shows that it has a better performance than the Hedonic Model, but failed to perform better than the Random Forest. This model has a MAE = 789,674 € and MAPE = 23,780%.

	Linear Regression	Random Forest	Artificial Neural Network
MAE	985,297	623,645	789,674
MAPE	30,628	19,386	23,780

Table 3 – Comparison of the Mean Absolute Error and Mean Absolute Percentage Error of the different models.



a) Hedonic Regression Model b) Random Forest Model c) Artificial Neural Network

Figure 3 – Scatterplots for the machine learning algorithms predictions.

Overall, the most expensive properties were the hardest to predict. There is a lack of data when it comes to the most valued households, since in the dataset there are only 68 properties that have a price

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per square meter superior to 10.000€, thus making it difficult for the model to learn how to predict the value of these properties. It is also important to mention that the results from the Artificial Neural Network could be inconsistent when repeating the process, since when this algorithm is initialized, the network weights are randomly assigned. Moreover, the shortage of predictive power by the ANN could come from the lack of data, since ANNs are known to better estimate when there is more data. Besides that, Neural Networks can also seem inconsistent when dealing with smaller samples of data, producing good training results by over-training the algorithm with the small samples, and producing worst results when dealing with the test data, even though its ability to deal with outliers. In figure 4, the MAPE is depicted according to the geographical distribution of the properties, for the test dataset. The areas in green are the areas that have less error, while red areas represent a higher error of estimation. As expected, the Hedonic Regression Model has a higher error for the estimation of each property, which can be seen from broader areas of red in Figure 4a). However, when comparing to the Random Forest (Figure 4b)) and Artificial Neural Network (Figure 4c)), it is clear that both these models have less broader areas of red in them, meaning that the estimates are more precise, mainly in the center of Lisbon, where properties are more difficult to estimate. This means that these models make better estimates of the price per square meter than the Hedonic Regression Model.

a) Hedonic Regression Model b) Random Forest Model c) Artificial Neural Network

Figure 4 – Heatmap of the Absolute Percentage Error of each property for the algorithms.

In order to corroborate what is being shown on the previous figure, Table 4 provides the number of properties observed and the Mean Absolute Percentage Error (MAPE) of the Random Forest Model, for each parish in Lisbon, along the years included in this study. The data in the table represents the data for the test set, which is unseen data for the algorithm. The most expansive locations, which are marked as red in Figure 4, are included in the Parishes that record some of the highest errors. These locations are where the most sightseeing activities happen or where housing facilities have the best conditions, thus raising the average price per square meter and making more difficult for the algorithm to predict the prices. As previously mentioned, the origin of such high errors could also come from the lack of information about properties in that area or from the disparity of prices between habitation residences in the area. In general, the MAPE for each Parish usually decreases over the years, which might imply the algorithm's ability to learn the importance of the time axis, when calculating the final price per square meter. Furthermore, some parishes don't have a MAPE for a determined year, resulting from the absence of information about real estate facilities, that could come from the train-test split or simply by the lack of data for that region

Parish	2017	2018	2019	2020	2021	2022

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	Obs.	MAPE	Obs.	MAPE	Obs.	MAPE	Obs.	MAPE	Obs.	MAPE	Obs.	MAPE
Ajuda	1	19,123	36	27,249	63	23,559	41	19,938	41	30,984	5	12,125
Alcantâra	6	8,829	62	23,037	50	19,476	67	12,498	46	16,579	0	------
Alvalade	7	21,376	103	15,265	95	15,645	81	19,384	85	14,984	11	7,498
Areeiro	11	35,487	68	22,639	58	18,82	81	23,062	49	21,85	6	13,165
Arroios	27	32,232	221	21,381	199	20,429	185	20,824	172	22,797	14	19,161
Avenidas Novas	20	15,653	138	22,525	126	18,002	116	17,443	146	12,405	14	13,208
Beato	2	3,799	39	21,643	33	18,49	33	15,092	40	17,723	7	12,979
Belém	11	19,461	50	26,564	73	14,964	54	16,981	61	18,524	2	18,797
Benfica	8	15,12	44	17,584	62	19,685	45	14,286	57	18,489	5	6,451
Campo de Ourique	9	20,433	101	18,479	98	25,505	77	19,304	90	20,033	7	23,217
Campolide	5	21,378	52	17,969	39	17,463	68	15,172	49	16,696	3	13,566
Carnide	4	48,813	30	12,572	20	15,54	16	14,576	27	13,636	1	12,452
Estrela	11	18,637	97	24,67	90	28,983	109	21,987	114	22,243	10	18,191
Lumiar	16	17,18	50	21,565	74	12,517	70	15,809	89	12,185	1	1,77
Marvila	3	20,116	60	19,471	89	18,071	42	13,178	59	13,899	5	21,224
Misericórdia	27	28,09	112	17,87	79	19,087	99	18,876	73	26,532	8	14,057
Parque das Nações	14	21,74	34	18,962	40	16,355	33	23,621	48	18,174	4	11,783
Penha de Franca	6	35,21	112	19,144	165	19,247	95	18,215	127	23,286	9	12,154
Santa Clara	6	14,62	10	11,691	23	20,932	19	15,207	15	12,613	0	------
Santa Maria dos Olivais	4	21,243	37	19,843	34	16,029	32	12,454	62	16,219	4	25,137
Santa Maria Maior	34	22,628	120	19,225	109	19,986	77	19,103	67	24,258	1	37,023
Santo António	26	18,807	103	21,791	116	17,368	87	17,234	91	16,325	15	19,846
São Domingos de Benfica	16	24,507	53	18,828	39	13,821	49	17,075	52	12,015	3	20,506
Sao Vicente	17	24,258	85	21,227	95	22,928	77	18,865	72	23,157	11	45,156

Table 4 – Comparison of the MAPE of the Test dataset for each Parish and Year.

The Random Forest presented greater predictive accuracy, and therefore this model would predict the price per square meter of each property more accurately. However, from the user's perspective, a Hedonic Model is easier to use and has a faster execution time when compared to the Random Forest and ANN, as both the Random Forest and ANN models are more complicated to use and have longer execution times but predict the price per square meter more accurately. Furthermore, the associations made by the Hedonic model can be easily interpreted, since it is a white-box system, while both the ANN and RF models are black-box systems and the interpretability of the associations made by the model may never be fully understood. In order to make better predictions, both the ANN and Random Forest need a larger dataset, that can become burdensome in some situations, as data availability for certain locations can be scarce. However, the ANN and Random Forest address some common problems for the Hedonic Model, such as the existing non-linearity for the most valued properties, and the ability to easily deal with outliers.

5. Conclusion

The purpose of this research is to determine the price of a real estate property using machine

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learning algorithms, as a tool to potentiate the existing physical data about the properties in Lisbon. Altogether, the findings of this study point out that the best scoring model is the Random Forest, followed by the Artificial Neural Network and finally, the Linear Regression/Hedonic Pricing Model. These results help confirm the Literature Review that was made, by proving that HPM's are outdated and that there are better outperforming Machine Learning algorithms to choose from. HPM's are widely used by real estate investors to predict the price of properties, despite its lack of predictive power when compared to other AI models, as the ever-growing amount of data that exists about real estate could be better applied if algorithms with enhanced capabilities are used. Despite everything, for future research, the models should be tested with more data and a geographical expansion of the properties studied should be made. A more complete dataset with more properties around the research area could improve the results of the machine learning models. Besides that, more variables about the housing facilities, such as number of bathrooms, existence of garden and the type of building materials could also be an improvement on the models, leveraging the predictive accuracy. The models used for this research can be very helpful in the real estate paradigm for real estate companies, as they can accurately and objectively assess the selling price or asking price of a dwelling, as well as estimate the valuation of certain features of a property. Moreover, these models can also be helpful for the real estate clients, as they can assess the existing opportunities, whether to make an investment or simply to buy a living home. Finally, using these algorithms, the real estate business can be improved, leaving subjectivity out of the equation and even an automated pipeline of commercialization could be created, reducing or closing the need for human intervention in the real estate markets. There is a lot of margin for improvement in this area, however making machines learn and predict for us is a huge step-up for mankind.

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Hedonic Model, Random Forest and Artificial Neural Network: Comparison for Real Estate Price Prediction in Lisbon

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Abstract

The real estate market is a complex industry with many variables affecting the prices of transactions. Formerly, understanding this market could be restricted by the lack of data or tools. However, with the current developments in the Artificial Intelligence field, three machine learning models (Hedonic Pricing Regression Model, Random Forest and Artificial Neural Network) are deployed and set side by side on the topic of predicting the price per square meter of housing properties, using real-estate data from Lisbon, Portugal. Overall, model evaluation reveals that the Random Forest was the best performing model, followed by the Artificial Neural Network and the Hedonic Model. These can predict the selling price per square meter accurately and learn the importance of certain features. Furthermore, the models implemented allow for users in the real estate paradigm to accurately and objectively assess the selling price or asking price of a dwelling, as well as estimate the valuation of certain features of a property.

Keywords

Real Estate, House Price Prediction, Machine Learning (ML), Hedonic Pricing Model (HPM), Random Forest (RF), Artificial Neural Network (ANN)

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